CYCLONES, STORM SURGE AND COASTAL PREPAREDNESS

COMPOUND CYCLONE HAZARD MAP → CYCLOGENESIS BOUNDARY → COASTAL WATER FIREWALL → IMD INFORMATION RAIL → COLOUR-STAGE MATRIX → WARNING-TO-EVACUATION CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00

COMPOUND CYCLONE HAZARD MAP

COMPOUND CYCLONE HAZARD MAP · STRUCTURE DEBRIS POWER TELECOM · RIVERINE FLASH PLUVIAL URBAN FLOOD · STORM SURGE + WAVES · COASTAL INUNDATION EROSION · TRANSPORT HEALTH WATER AND LIVELIHOODS

WIND → structure debris power telecom

RAIN → riverine flash pluvial urban flood → STORM SURGE + WAVES → coastal inundation erosion → CASCADE → transport health water and livelihoods

HAZARD / RISK / RESILIENCE CONCEPT
- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

INSTITUTIONAL OR COMMUNITY RESPONSE
- STORM SURGE + WAVES → coastal inundation erosion
- CASCADE → transport health water and livelihoods

CAPACITY / STATUS / EVIDENCE LIMIT
- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE + WAVES → coastal inundation erosion.

01

STAGE 01

CYCLOGENESIS BOUNDARY

CYCLOGENESIS BOUNDARY · WARM WATER + INSTABILITY + MOISTURE · CORIOLIS + DISTURBANCE + LIMITED SHEAR · CYCLONIC SYSTEM · DM FOCUS · EXPOSURE WARNING ACTION CONTINUITY

WARM WATER + INSTABILITY + MOISTURE → CORIOLIS + DISTURBANCE + LIMITED SHEAR → CYCLONIC SYSTEM → DM FOCUS → exposure warning action continuity

HAZARD / ACTOR / INSTRUMENT
- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

DECISIVE RISK OR MANDATE DISTINCTION
- CYCLONIC SYSTEM
- DM FOCUS → exposure warning action continuity

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CYCLONIC SYSTEM; DM FOCUS → exposure warning action continuity.

02

STAGE 02

COASTAL WATER FIREWALL

COASTAL WATER FIREWALL · STORM SURGE · CYCLONE WIND AND PRESSURE · ASTRONOMICAL GRAVITATIONAL CYCLE · SUDDEN WATER DISPLACEMENT · TOTAL WATER LEVEL · INTERACTING BUT DISTINCT COMPONENTS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
STORM SURGE → cyclone wind and pressure	TSUNAMI → sudden water displacement	STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle	TOTAL WATER LEVEL → interacting but distinct components	TIDE → astronomical gravitational cycle

HAZARD / ACTOR / INSTRUMENT
- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION
- TSUNAMI → sudden water displacement
- TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOTAL WATER LEVEL → interacting but distinct components.

03

STAGE 03

IMD INFORMATION RAIL

IMD INFORMATION RAIL · 1 OBSERVE · 2 CLASSIFY · FORECAST TRACK AND INTENSITY · 3 ISSUE IMPACT AND HAZARD INFORMATION · 4 UPDATE AS UNCERTAINTY CHANGES

HAZARD / RISK / RESILIENCE CONCEPT
- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 ISSUE IMPACT AND HAZARD INFORMATION
- 4 UPDATE AS UNCERTAINTY CHANGES

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE IMPACT AND HAZARD INFORMATION; 4 UPDATE AS UNCERTAINTY CHANGES.

04

STAGE 04

COLOUR-STAGE MATRIX

COLOUR-STAGE MATRIX · ACTION LEVEL · BULLETIN STAGE · LEAD-TIME SEQUENCE · REQUIRE LOCAL ACTION PROTOCOL

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
GREEN / YELLOW / ORANGE / RED → ACTION LEVEL	COLOUR != LEAD-TIME SEQUENCE	GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE	BOTH → require local action protocol	WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

HAZARD / ACTOR / INSTRUMENT
- GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
- WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

DECISIVE RISK OR MANDATE DISTINCTION
- COLOUR != LEAD-TIME SEQUENCE
- BOTH → require local action protocol

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
- WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE.

05

STAGE 05

WARNING-TO-EVACUATION CHAIN

WARNING-TO-EVACUATION CHAIN · OFFICIAL FORECAST · AREA DECISION · TRANSPORT + ROUTE + LIVESTOCK · ASSET PLAN · ACCESSIBLE SHELTER + REGISTRATION · ALL CLEAR · COMPETENT AUTHORITY

OFFICIAL FORECAST → AREA DECISION → TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN → ACCESSIBLE SHELTER + REGISTRATION → ALL CLEAR / RETURN → COMPETENT AUTHORITY

RISK CONDITION / INPUT
- OFFICIAL FORECAST → AREA DECISION
- TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- ACCESSIBLE SHELTER + REGISTRATION
- ALL CLEAR / RETURN → COMPETENT AUTHORITY

LAST-MILE / INCLUSION / OUTCOME LIMIT
- OFFICIAL FORECAST → AREA DECISION
- TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN.

06

STAGE 06

SHELTER READINESS TEST

SHELTER READINESS TEST · SAFE SITE AND ACCESS? · WATER SANITATION POWER ACCESSIBILITY? · TRAINED MANAGEMENT AND MAINTENANCE? · ASSET EXISTS BUT READINESS UNPROVEN

SAFE SITE AND ACCESS?

WATER SANITATION POWER ACCESSIBILITY? → TRAINED MANAGEMENT AND MAINTENANCE? → NO → ASSET EXISTS BUT READINESS UNPROVEN

HAZARD / RISK / RESILIENCE CONCEPT
- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

INSTITUTIONAL OR COMMUNITY RESPONSE
- TRAINED MANAGEMENT AND MAINTENANCE?
- NO → ASSET EXISTS BUT READINESS UNPROVEN

CAPACITY / STATUS / EVIDENCE LIMIT
- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAINED MANAGEMENT AND MAINTENANCE?; NO → ASSET EXISTS BUT READINESS UNPROVEN.

07

STAGE 07

CRITICAL-SERVICE WEB

CRITICAL-SERVICE WEB · SUPPLY CHAIN · EOC / HOSPITAL / RESCUE · POWER / WATER / TELECOM

EOC / HOSPITAL / RESCUE

POWER / WATER / TELECOM → ROAD / PORT / SUPPLY CHAIN → REDUNDANCY → ASSESS → RESTORE

HAZARD / RISK / RESILIENCE CONCEPT
- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE
- ROAD / PORT / SUPPLY CHAIN
- REDUNDANCY → ASSESS → RESTORE

CAPACITY / STATUS / EVIDENCE LIMIT
- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROAD / PORT / SUPPLY CHAIN; REDUNDANCY → ASSESS → RESTORE.

08

STAGE 08

HOUSING-TO-ECOSYSTEM PORTFOLIO

HOUSING-TO-ECOSYSTEM PORTFOLIO · SITING CODE COMPLIANCE MAINTENANCE · LIFELINE HARDENING AND REDUNDANCY · MANGROVE DUNE WETLAND PROTECTION · LAYERED DEFENCE WITH RESIDUAL RISK

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
HOUSING → siting code compliance maintenance	ECOSYSTEM → mangrove dune wetland protection	HOUSING → siting code compliance maintenance
INFRASTRUCTURE → lifeline hardening and redundancy	RULE → layered defence with residual risk	INFRASTRUCTURE → lifeline hardening and redundancy

HAZARD / ACTOR / INSTRUMENT
- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

DECISIVE RISK OR MANDATE DISTINCTION
- ECOSYSTEM → mangrove dune wetland protection
- RULE → layered defence with residual risk

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ECOSYSTEM → mangrove dune wetland protection; RULE → layered defence with residual risk.

09

STAGE 09

NCRMP STATUS LADDER

NCRMP STATUS LADDER · PROJECT APPROVAL · ASSET CREATION · PHASE COMPLETION · LOCAL HAND-OFF · OPERATION + MAINTENANCE + DRILLS · CURRENT READINESS · NEEDS SEPARATE EVIDENCE

PROJECT APPROVAL → ASSET CREATION

PHASE COMPLETION → STATE / LOCAL HAND-OFF → OPERATION + MAINTENANCE + DRILLS → CURRENT READINESS → needs separate evidence

RISK CONDITION / INPUT
- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- OPERATION + MAINTENANCE + DRILLS
- CURRENT READINESS → needs separate evidence

LAST-MILE / INCLUSION / OUTCOME LIMIT
- MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,..
- PROJECT APPROVAL → ASSET CREATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,.

10

STAGE 10

LIVELIHOOD RECOVERY LOOP

LIVELIHOOD RECOVERY LOOP · BOAT GEAR LIVESTOCK AND MARKET ACTION · RELIEF + SOCIAL PROTECTION · SAFER CONDITIONS · BUILD BACK BETTER · REDUCE FUTURE EXPOSURE · CLOSE DISTINCTION · CYCLONE CATEGORY IS NOT STORM-SURGE HEIGHT; FORECAST LANDFALL IS

WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION

IMPACT → RELIEF + SOCIAL PROTECTION → REPAIR / REOPEN → SAFER CONDITIONS → BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

HAZARD / RISK / RESILIENCE CONCEPT
- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION
- IMPACT → RELIEF + SOCIAL PROTECTION

INSTITUTIONAL OR COMMUNITY RESPONSE
- REPAIR / REOPEN → SAFER CONDITIONS
- BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

CAPACITY / STATUS / EVIDENCE LIMIT
- CLOSE DISTINCTION: Cyclone category is not storm-surge height; forecast landfall is not...
- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Cyclone category is not storm-surge height; forecast landfall is not.

11

STAGE 11

CYCLONE ANSWER SPINE

CYCLONE ANSWER SPINE · DEFINE COMPOUND HAZARD · SEPARATE SURGE TIDE TSUNAMI · MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE · CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS · ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION · EVIDENCE LIMIT · USE IMD TERMINOLOGY AND TIMESTAMPED

DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI → MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE → CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS → ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped

HAZARD / RISK / RESILIENCE CONCEPT
- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI
- MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE

INSTITUTIONAL OR COMMUNITY RESPONSE
- CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS
- ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION

CAPACITY / STATUS / EVIDENCE LIMIT
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped...
- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · MULTI-SECTORAL PREMISE, EXPLICITLY STATED · NDMA'S 2008 CYCLONE · NCRMP'S INSTITUTIONAL STRUCTURE · NDMA (UNDER MHA) IMPLEMENTS IN · ASYMMETRIC COASTAL EXPOSURE, PRECISELY QUANTIFIED IN THE SOURCE · 1994 ODISHA CYCLONE AS A CASCADING-HAZARD CASE · AIRCRAFT PROBING OF CYCLONE (APC) AS AN UNDER-INSTITUTIONALISED

OPTIONAL RISK / RESILIENCE DEPTH
- Multi-sectoral premise, explicitly stated : NDMA's 2008 cyclone
- NCRMP's institutional structure : NDMA (under MHA) implements in
- Asymmetric coastal exposure, precisely quantified in the source :
- CAUTION: 1994 Odisha cyclone as a cascading-hazard case : cyclone

MANDATE, WARNING OR OUTCOME LIMIT
- CAUTION: Aircraft Probing of Cyclone (APC) as an under-institutionalised
- CAUTION: VisionIAS names "a lack of a fully automated and state-of-the-art
- CAUTION: VisionIAS's "lack of grass root level participation in disaster
- CAUTION: Category-based investment prioritisation is itself a trade-off.

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- CAUTION: Saline-embankment/bio-shield maintenance versus new-construction
- NOT: The "NCRMP exists, therefore coastal States are protected"
- CAUTION: Network fragmentation across agencies : the explicit call for
- NCRMP's blended financing (NDMA/MHA-led, State-coordinated,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels